

**AP Question Paper 2018****Class 12 Chemistry, Paper - II****Time:3 Hours****Max.Marks:60****Note: Read the following instructions carefully:**

- 1) Answer all the questions of Section-‘A’. Answer any six questions from Section-‘B’ and any two questions from Section-‘C’.**
- 2) In Section-‘A’, Questions from Sr. Nos. 1 to 10 are of “very short answer type”. Each question carries two marks. Every answer may be limited to two to three sentences. Answer all the questions at one place in the same order.**
- 3) In Section-‘B’, Questions from Sr. Nos. 11 to 18 are of “Short answer type”. Each questions carries four marks. Every answer may be limited to 75 words.**
- 4) In Section-‘C’, Questions from Sr.Nos.19 to 21 are of” Long answer type”. Each questions carries eight marks. Every answer may be limited to 300 words.**
- 5) Draw labelled diagram, wherever necessary for questions in section-‘B’ and ‘C’.**

**SECTION- A** **$10 \times 2 = 20$** **Note: Answer all the questions:****1. Define Osmotic Pressure.**

Ans:- Osmotic Pressure is the pressure that would have to be applied to a pure solvent to prevent it from passing into a given solution by osmosis, it is often used to express the concentration of the solution.

**2. What are antibiotics? Give example.**

Ans:-An antibiotic is a type of antimicrobial substance active against bacteria and is the most important type of antibacterial agent for fighting bacterial infections. Antibiotic medications are widely used in the treatment and prevention of such infections. They may either kill or inhibit the growth of bacteria.

Ex:-Penicillin

**3. Write the difference between a soap and a synthetic detergent.**

Ans:-1) soap are prepared from vegetables and oil but detergent are prepared from the hydrocarbons obtained from petroleum.

2) soap are bio degradable but detergent are non biodegradable

3) soap do not cause water pollution but detergent cause water pollution

**4. State Faraday's first law of electrolysis.**

Ans:- Faraday's First Law of Electrolysis states that the chemical deposition due to the flow of current through an electrolyte is directly proportional to the quantity of electricity (coulombs) passed through it.

i.e. mass of chemical deposition:

$$m \propto \text{Quantity of electricity, } Q \Rightarrow m = Z \cdot Q$$

Where,

Z is a constant of proportionality and is known as electro-chemical equivalent of the substance.

**5. Give the composition of the following alloys:**

- a) Brass    b) German silver

Ans: (a) Brass: alloy of Copper and Zinc. (Cu + Zn)

(b) German silver: alloy of Copper, Zinc and Nickel.

(Cu + Zn + Ni)

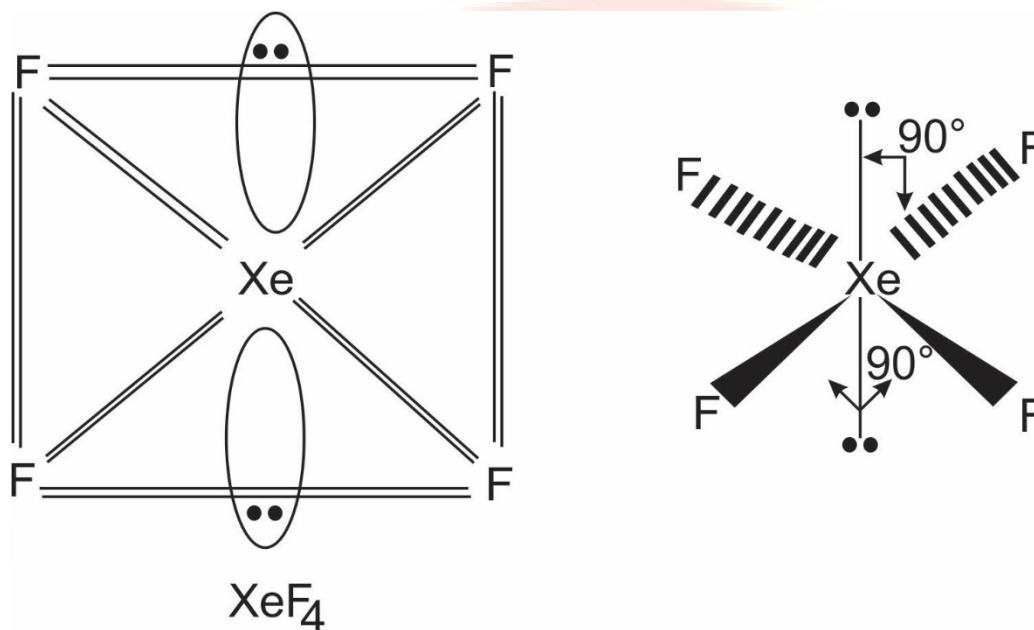
**6. List out any two uses of Neon.**

Ans:-The two uses of neon are :-

- It is also used to make high-voltage indicators, lightning arrestors, wave meter tubes, and TV tubes.
- In making gas lasers neon and helium is used.

**7. Write the structures of XeF<sub>4</sub>.**

Ans:- The VSEPR structure of  $\text{XeF}_4$  is square planar. It is an octahedral but because of the lone pairs, it dictate there 6 Domains around the central atom and the VSEPR theory states any  $\text{AX}_4\text{E}_2$  specie with 2 lone pairs is square planar.



**8. Why  $\text{Zn}^{2+}$  is diamagnetic whereas  $\text{Cr}^{2+}$  is paramagnetic?**

Ans:  $\text{Zn}^{2+}$  having a completely occupied d orbital with d10 electronic configuration whereas  $\text{Cr}^{2+}$  has d9 electronic configuration having paramagnetic complex.

**9. What is biodegradable polymer? Give one example.**

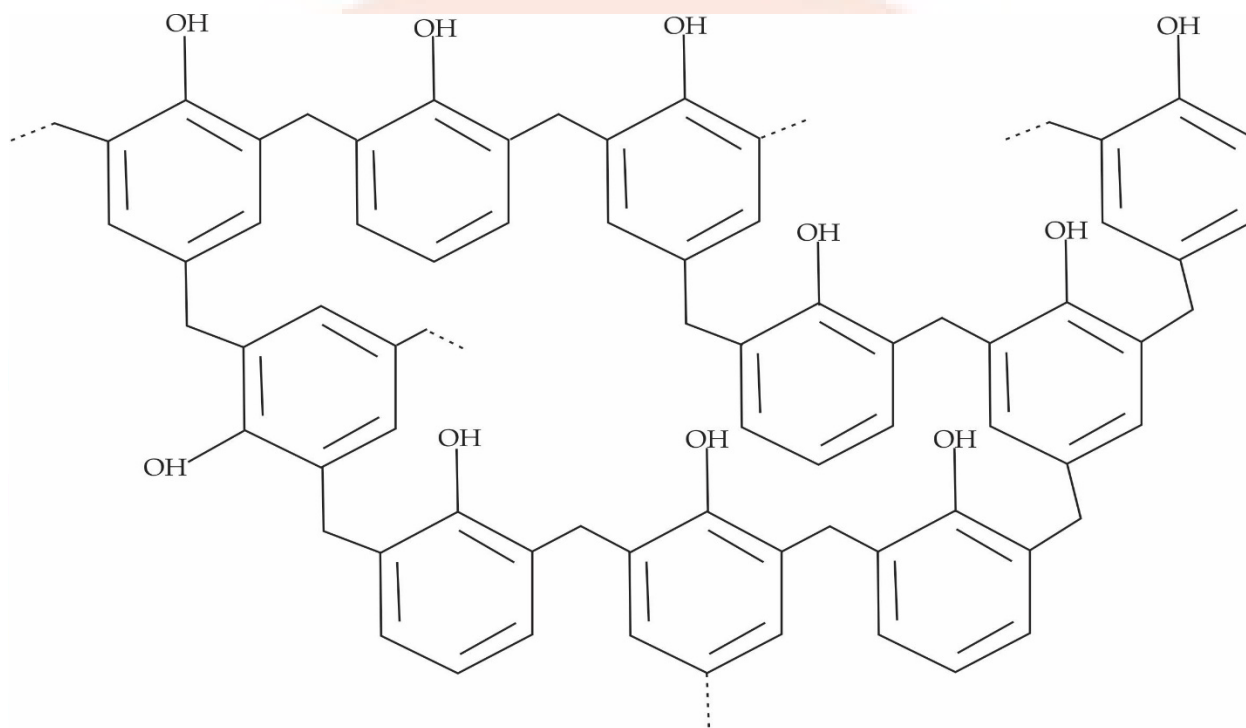
Ans: Biodegradable polymers are a special class of polymer that breaks down after its intended purpose by bacterial decomposition process to result in natural byproducts such as gases, water, biomass, and inorganic salts

Example:-Polyhydroxy Butyrate(PHB)

**10. Write the names of the monomers of the following polymers:**

a) Bakelite      b) Nylon-6,6

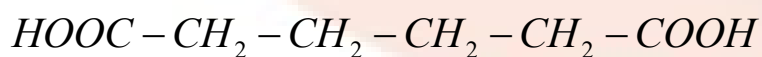
Ans:- i) Phenol, Formldehyde



ii) hexamethylenediamine and adipic acid



Hexamethylenediamine



adipic acid

**SECTION – B****(6 × 4 = 24)****Note : Answer any six questions.****11. What is doping? Explain n-type and p-type semiconductors.**

Ans:- A doped semiconductor (as opposed to an intrinsic semiconductor) is one that has had specific impurity atoms introduced into it to shrink the band gap.

An N-type :- semiconductor is one that has been doped with a pentavalent element (that is, five valence electrons). This additional electron (silicon has four) shrinks the band gap by expanding the conduction band (the higher energy levels).

A P-type:- semiconductor is exactly the opposite: a semiconductor doped with a trivalent element (three valence electrons). This "donates holes" to the structure, and shrinks the band gap by expanding the valence band (lower energy levels).

**12. Explain the purification of Sulphide ore by froth flotation method.**

Ans:- This method is based on the principle of the difference in the wetting properties of the ore and the gangue particles with the water and oil. This method is used for that metals whose ore gets preferably wetted by the oil and the gangue by the water. The sulphide ores preferably wetted by the oil and the gangue with the water. Hence this method is suitable for the concentration of the sulphide ore.

**13. State Raoult's law. The vapour pressure of pure Benzene at a certain temperature is 0.850 bar. A non-Volatile, non-electrolyte solid weighing 0.5g when added to 39.0g of benzene. Vapour pressure of the solution, then is 0.845 bar. What is the molar mass of the solid substance?**

Ans:- Raoult's law states that in a solution, the vapour pressure of a component at a given temperature is equal to the mole fraction of that component in the solution multiplied by the vapour pressure of that component in the pure state.

The total vapour pressure  $P$  of a solution containing two components A and B is

$$P_A = P_A^0 \times X_A$$

$$P_B = P_B^0 \times X_B$$

$$P = P_A + P_B = P_A^0 X_A + P_B^0 X_B$$

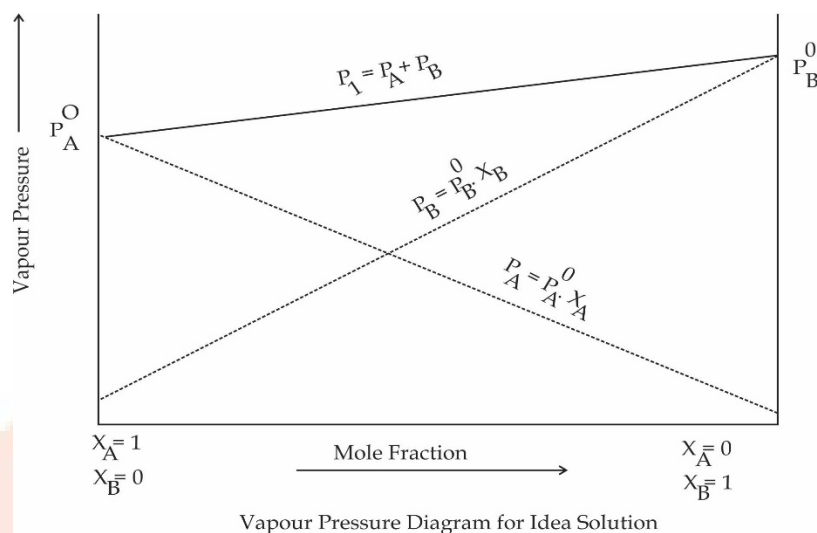
As we know that

$$X_A + X_B = 1$$

$$X_A = (1 - X_B)$$

$$P = (1 - X_B)P_A^0 + P_B^0 X_B$$

$$P = (P_B^0 - P_A^0)X_B + P_A^0$$



$$p^0 = 0.850 \text{ bar}$$

$$p = 0.845 \text{ bar}$$

$$w_B = 0.5 \text{ g}$$

$$w_A = 39.0 \text{ g}$$

$$M_B = ?$$

Applying,

$$\begin{aligned} M_B &= \frac{w_B M_A}{w_b \left( \frac{p^0 - p}{p^0} \right)} \\ &= \frac{0.5 \times 78}{39.0 \times \left( \frac{0.850 - 0.845}{0.850} \right)} \\ &= \frac{0.5 \times 2 \times 0.850}{0.005} \\ &= 170 \text{ g mol}^{-1} \end{aligned}$$

Hence, molar mass of the solid substance = 170 g/mol



#### **14. Describe the purification of the colloidal solutions by Dialysis method with a neat diagram.**

Ans: Purification:- the process of reducing the impurities of the electrolytes to the minimum required level is known as purification of colloidal solution.

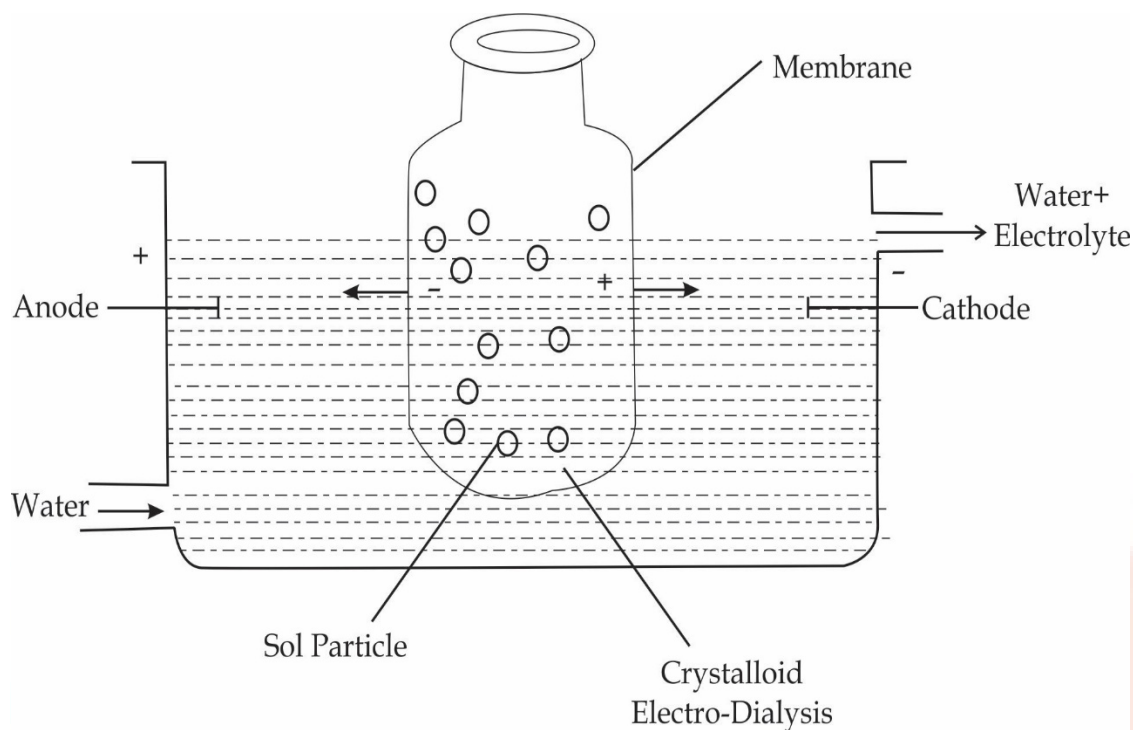
Dialysis:-The process of separating the particles of colloids from those of crystalloids by diffusion of the mixture through a parchment or an animal membrane is called dialysis.

Process:- The apparatus used is called a dialyser. A cellophane sheet or parchment paper is turned into a bag with a funnel tied in the mouth of the bag, for the addition of impure sol. The impure sol is filled into the bag which is then suspended into a vessel containing distilled water.

Electro dialysis:- In this method electric field is applied using the metal electrodes, and the diffusion of the given mixture through using parchment or an animal membrane is known as Electro dialysis.

Ultrafiltration:- First of all filter paper is treated with colodion or gelatin solution followed by hardening by dipping into formaldehyde. The colloid in solution used is 4% solution of nitrocellulose in a mixture of alcohol and ether. Due to this pore size of filter paper is reduced. This filter paper so obtained is called an ultra filter paper.

→ and purification by this called ultrafiltration.



**15. Write the equations for reactions of chlorine with the following.**

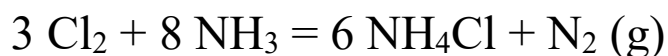
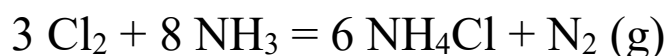
a) cold and dilute NaOH      b) Excess  $\text{NH}_3$

c)  $\text{Ca}(\text{OH})_2$       d)  $\text{Na}_2\text{S}_2\text{O}_3$

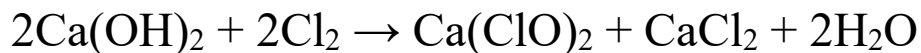
Ans:-a) The reaction between chlorine and cold dilute sodium hydroxide solution gives  $\text{NaClO}_3$  sodium chlorate(I).



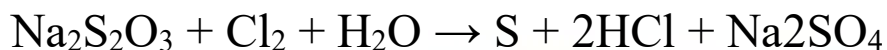
b) The reaction between chlorine and  $\text{NH}_3$  gives white fumes of  $\text{NH}_4\text{Cl}$ , with excess of ammonia it gives nitrogen chloride.



c) Calcium hydroxide reacts with chlorine



d) Sodium thiosulfate react with chlorine and water



### 16. Explain Werner's theory of Co-ordination compounds.

Ans:- Werner's theory coordination compounds:

- Two types of valency are exhibited by the central metal atom. Example- primary valency and the secondary valency.
- Secondary valency are fixed in number in the every cental metal atom. Hence it has a fixed coordinate number.
- Both primary and secondary valencies are satisfied by the metal atom, in which primary valency is satisfied by negative ion and whereas secondary valency are satisfied with neutral molecules.
- Definite geometry of coordinated compound is caused by the secondary valency which is always directed towards fixed position in space.

### 17. What are hormones? Give one example for each of the following:

a) Steroid hormones

b) Polypeptide hormones

### c) Amino acid derivatives

Ans:-The molecules produced by the endocrine system that send messages to different parts of body regulating body's process like hunger, blood pressure and sexual desire are called hormones.

(A)Sex hormones (androgens, estrogens, progesterone):-Steroid hormones are fat-soluble molecules derived from cholesterol. They are produced by certain endocrine organs and glands and released into the bloodstream to reach target cells.

(B)Insulin, Prolactin:-Prolactin, a multifunctional hormone, is involved in regulating insulin sensitivity and blood glucose stability in experimental studies. It helps regulate growth of insulin-producing cells in the pancreas, and is also involved in lactation, reproduction, metabolism, immune regulation and water balance (osmoregulation).

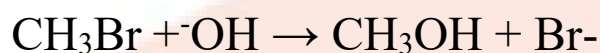
(C)Epinephrine and norepinephrine:-Epinephrine and norepinephrine are released by the adrenal medulla and nervous system respectively. They are the flight/fight hormones that are released when the body is under extreme stress. During stress, much of the body's energy is used to combat imminent danger.

### 18.Explain the mechanism of SN<sub>2</sub> reaction with example.

Ans:- SN<sub>2</sub> (substitution, nucleophilic, bimolecular) takes place in a single step without intermediates when a nucleophile reacts with the substrate(e.g. alkyl halide) from a direction opposite

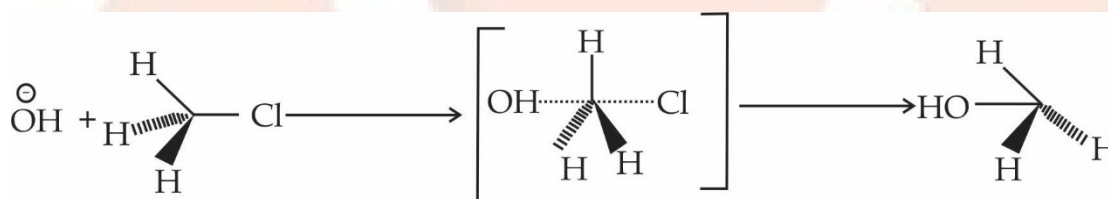
opposite to the leaving group.

The reaction between methyl bromide and hydroxide ion to yield methanol follows second order kinetics; that is, the rate depends upon the concentrations of both reactants :



$$\text{rate} = K [\text{CH}_3\text{Br}] [\text{OH}^-]$$

The reaction is believed to take place as shown:



## SECTION – C

(2 × 18 = 16)

**Note : Answer ANY TWO questions.**

19. a) What are Galvanic cells? Explain the working of Galvanic cells with one example.

Ans:- :- Galvanic cells – An electrical energy from redox reaction take place within the cell via electrochemical cell. This cell is known as galvanic cell.

Working of a galvanic cell:

- It involves a chemical reaction which makes the electric energy.
- In the redox reaction, galvanic cell used the energy transfer

between the electrons which converts the chemical energy into electric energy.

- Galvanic cell is used to separate the flow of electrons. This flow of electrons is basically known as current.
- This galvanic cell consists of two metals, these two metals form the anode and other form the cathode.

Example : two metals involved in chemical reaction in which zinc end up losing two electrons ,and these electrons is taken by the copper. If these two metals are placed in separate container the would be connected by the conducting wire , a electric current is generated in them. In this case , two solutions are not mixed together directly but can be joined by the salt bridge.

b)Write the differences between order and molecularity of a reaction.

Ans:-1.Order need not be a whole number. It can be fractional or zero. Molecularity is always a whole number.

2. Order can be determined experimentally only. Molecularity can be calculated by adding the no. of molecules in the slowest step (directly by looking at the balanced equation).

3. Order is applicable to simple + elementary reactions. Molecularity is applicable to simple reactions. No significance in determining molecularity of a complex reaction.

4. Order cannot be determined by looking at the stoichiometric coefficients. Molecularity can be determined by looking at the stoichiometric

**20. a) How is ammonia manufactured by Haber's Process? Explain.**

**Ans.** Ammonia is synthesized by the help of haber's process from nitrogen and hydrogen. Haber's process combines nitrogen from air with hydrogen and formed the ammonia. It is the exothermic and reversible reaction.



The constituents of ammonia  $\text{N}_2$  and  $\text{H}_2$  combine in a ratio of 1:3.

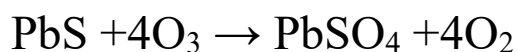
The conditions for the production of ammonia are:

- Pressure of approx 200atm
- Temperature of approx. 700K
- Catalyst such as iron oxide

b) How does ozone react with the following:

i)  $\text{PbS}$       ii)  $\text{KI}$       iii)  $\text{C}_2\text{H}_4$       iv)  $\text{NO}$

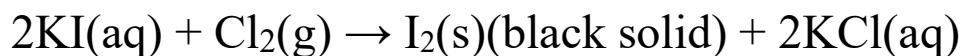
**Ans:-** i)  $\text{PbS}$ :- Lead Sulfide react with ozone to produce lead sulphate and oxygen at room temperature.



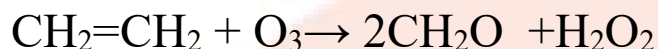
ii)  $\text{KI}$ :-Potassium Iodide reacts with chlorine to give iodine by



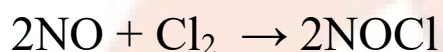
oxidizing iodide as chlorine is more stronger than KI.



iii)  $\text{C}_2\text{H}_4$ :- Ethylene reacts with ozone to produce ozonide which is decomposed by Zn dust and water to form formaldehyde.



iv) NO:-Chlorine reacts with NO to give nitrosyl chloride.

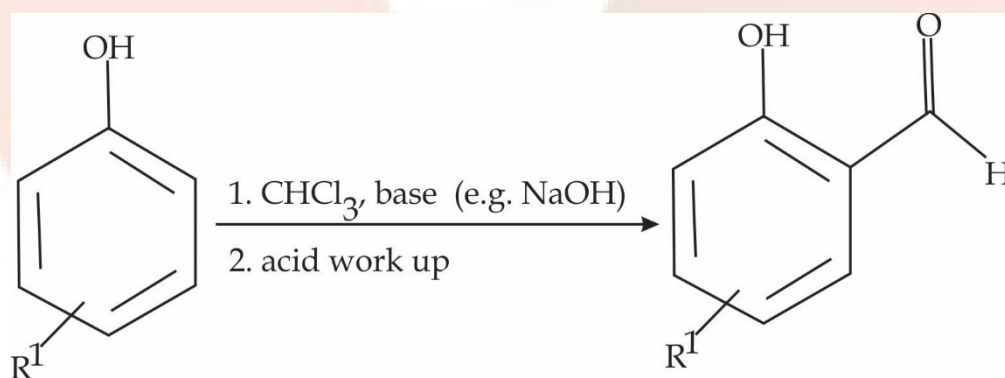


## 21. Explain the following reactions with suitable example:

a) Reimer-Tiemann reaction

Ans:- It is a type of substitution reaction. It is used for ortho-formylation of  $\text{C}_6\text{H}_5\text{OH}$ .

Example: Conversion of phenol to Salicylaldehyde.

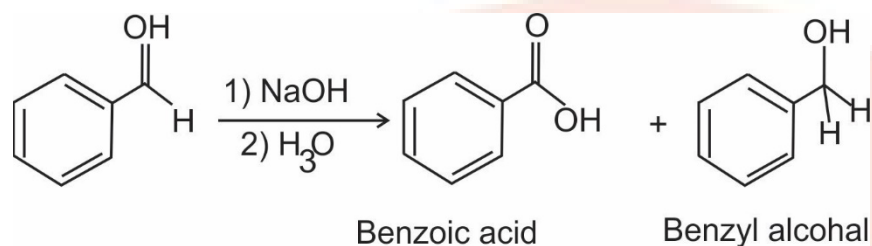


b) Cannizzaro reaction:-

Cannizzaro reaction is a reaction in which two molecules of an aldehyde are reacted to produce a primary alcohol and a carboxylic acid using a hydroxide base.

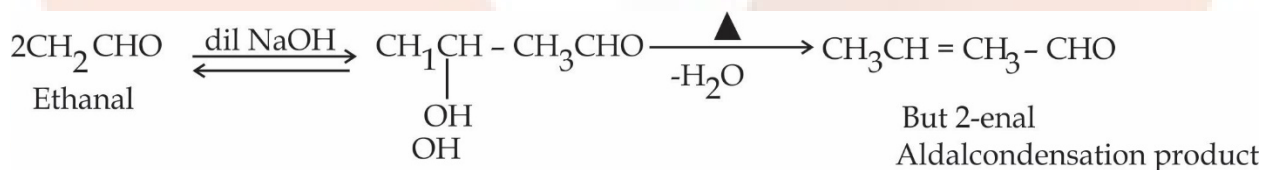


When two molecules of aldehyde is treated with sodium hydroxide, a reaction occurs in which one mole of aldehyde is oxidized (giving carboxylic acid) while the other mole of aldehyde is reduced (giving primary alcohol).



### c) Aldol Condensation

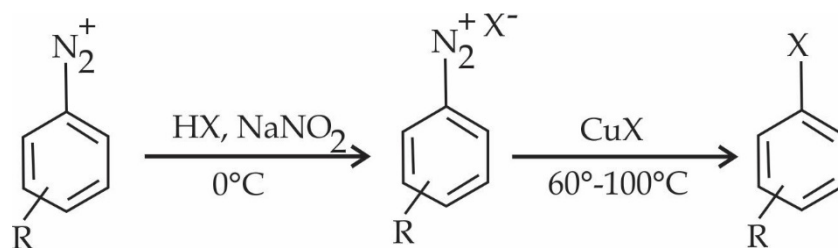
Ans: When aldehydes and ketones having at least one  $\alpha$ -hydrogen are treated with dilute alkali (which act as a catalyst) they form  $\beta$ -hydroxyaldehydes (aldol) or  $\beta$ -hydroxyketones (ketol) respectively. This reaction is known as aldol condensation.



### d) Sandmeyer Reaction:-

Ans:- In Sandmeyer's reaction, a benzeneamine or an Aniline is treated with sodium nitrite ( $\text{NaNO}_2$ ) in the presence of  $\text{HCl}$  or any suitable Halogen acids at  $273.15\text{K}$ .

This reaction gives a diazonium salt of that halogen, this diazonium salt is then again reacted with cuprous halide which results in formation of a Haloarene.



$\text{X} = \text{CN}, \text{Br}, \text{Cl}, \text{SO}_3\text{H}$